

- DNA fingerprinting, also known as DNA profiling, is a method used to identify individuals based on their unique DNA characteristics. This technique has become an essential tool in forensic science, allowing for the identification of individuals involved in crimes, the resolution of paternity and immigration disputes, and the identification of bodies in disaster situations.
- DNA fingerprinting is a method used to identify individuals based on their unique DNA characteristics. This technique, developed by Sir Alec Jeffery in 1984, utilizes the small portion of DNA that varies from person to person. In India, pioneers of this technology include Dr. V.K. Kashyap and Dr. Lal Ji Singh.

Principle of DNA fingerprinting

- Human DNA is largely (99.9%) identical across all individuals, but a fraction (0.1%) of it exhibits variations known as polymorphisms.
- It is these differences in sequence of DNA which make every individual unique in their phenotypic appearance.
- If one aims to find out genetic differences between two individuals or among individuals of a population, sequencing the DNA every time would be a daunting and expensive task.
- Imagine trying to compare two sets of 3×10^6 base pairs. DNA fingerprinting is a very quick way to compare the DNA sequences of any two individuals.

Repetitive DNA

- DNA fingerprinting involves identifying differences in some specific regions in DNA sequence called as repetitive DNA, because in these sequences, a small stretch of DNA is repeated many times.
- These repetitive DNA are separated from bulk genomic DNA as different peaks during density gradient centrifugation.
- These sequences normally do not code for any proteins.
- These sequences form a large portion of human genome.
- These sequence show high degree of polymorphism and form the basis of DNA fingerprinting
- These sequence show high degree of polymorphism (variation) in population and homologous chromosomes of an individual
- These sequences forms the basis of DNA fingerprinting
- Since DNA from every tissue (such as blood, hair-follicle, skin, bone, saliva, sperm etc.), from an individual show the same degree of polymorphism, they become very useful identification tool in forensic applications.

- Further, as the polymorphisms are inheritable from parents to children, DNA fingerprinting is the basis of paternity testing, in case of disputes.
- As polymorphism in DNA sequence is the basis of genetic mapping of human genome as well as of DNA fingerprinting, it is essential that we understand what DNA polymorphism means in simple terms.

Satellite DNA

- Depending on base composition (A : T rich or G : C rich), length of segment, and number of repetitive units, the satellite DNA is classified into many categories, such as micro-satellites, mini-satellites etc.

Micro-satellites :

- It has 1 to 6 bp repetition
- It is also called SSR (Simple Sequence Repeats)

Mini-satellites :

- It has 11 to 60 bp repetition
- It is also called VNTR (Variable Number of Tandem Repeats)

VNTR :-

- The VNTR belongs to a class of satellite DNA referred to as mini-satellite
- A small DNA sequence is arranged tandemly in many copy numbers
- The copy number varies from chromosome to chromosome in an individual
- The numbers of repeat show very high degree of polymorphism
- As a result the size of VNTR varies in size from 0.1 to 20 kb (100 to 20,000)
- The repetition number of VNTR is vary in homologous chromosome of an individual

DNA Fingerprinting Steps

1. **DNA Extraction:** This initial step involves isolating DNA from cells through a process known as cell lysis. In cases where DNA content is scarce, it can be amplified using Polymerase Chain Reaction (PCR), a method that enables the multiplication of DNA fragments to sufficient quantities for analysis.
2. **Digestion Using Restriction Enzymes:** Special enzymes called restriction endonucleases, cut the DNA at specific sequences. This produces DNA fragments of varying lengths.

3. **Gel Electrophoresis:** This technique separates DNA fragments based on their size and charge through the application of an electrical current across a gel matrix, usually made from agarose. DNA fragments are negatively charged due to their phosphate backbone and move towards the positive electrode, with smaller fragments moving faster and thus separating according to size. The process effectively sorts the DNA fragments, facilitating their analysis.
4. **Southern Transfer/Southern Blotting:** Named after its inventor, Edward Southern, this step involves transferring the separated DNA fragments from the gel to a more durable nylon membrane. The DNA is first denatured into single strands and then blotted onto the membrane, where they are fixed in place.
5. **Hybridization:** To identify specific sequences within the DNA fragments, such as Variable Number Tandem Repeats (VNTRs), single-stranded DNA probes labeled with radioactive phosphorus-32 are used. These probes bind to complementary DNA sequences on the fragments, marking the locations of interest.
6. **Autoradiography:** The nylon membrane, now with the bound probes, is exposed to X-ray film. The radioactive probes create specific bands on the film, revealing the presence and pattern of the VNTRs. This pattern, known as Restricted Fragment Length Polymorphism (RFLP), varies significantly among individuals, allowing for the identification of unique DNA signatures.
7. The probability of two unrelated individuals sharing the same DNA fingerprint is extremely low, making this technique a powerful tool for identification purposes. In India, the Center for DNA Fingerprinting and Diagnosis (CDFD) in Hyderabad is a key institution for DNA fingerprinting research and applications.

DNA Fingerprinting Applications

1. **Paternity Testing:** This is a common use of DNA fingerprinting, where the DNA profiles of a child, the mother, and potential fathers are compared to establish biological relationships.
2. **Criminal Identification:** In forensic science, DNA fingerprinting has become indispensable for linking suspects to crime scenes, especially in serious cases like murders and rapes. By comparing DNA fingerprints obtained from crime scenes with those of suspects, investigators can identify the perpetrator with a high degree of certainty.